# Chapter 2: Vector Spaces

Exercise Set Part 1 of 5

Show your reasoning and row operations on separate paper. Use exact arithmetic. All vector spaces are over the real numbers. Unless stated otherwise, solve by hand; software may be used to check results.

## A. Vector spaces and subspaces

1. For each subset of $\mathbb{R}^3$, decide whether it is a subspace under the usual operations. Justify your answer using closure, or give a counterexample. For each subspace, give a spanning set.

$$S_1 = \{(x, y, z)^T : x + 2y - z = 0\},$$
$$S_2 = \{(x, y, z)^T : x + 2y - z = 1\},$$
$$S_3 = \{(x, y, z)^T : xy = 0\}.$$

2. Let $P = \{(x, y, 0)^T : x, y \in \mathbb{R}\}$ and $L = \{t(1, 0, 1)^T : t \in \mathbb{R}\}$. Find $P \cap L$. Is $P \cup L$ a subspace? Justify your answer. Repeat both questions with $L_0 = \{t(1, 2, 0)^T : t \in \mathbb{R}\}$ in place of $L$.

3. Let $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$. Describe $C(D)$ by an equation relating its three components. Determine whether $\mathbf{b} = (2, -1, 1)^T$ and $\mathbf{c} = (2, -1, 0)^T$ belong to $C(D)$. Solve $D\mathbf{x} = \mathbf{b}$ if possible.

4. Prove that the nullspace of any real $m \times n$ matrix is a subspace of $\mathbb{R}^n$. If $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{b} \neq \mathbf{0}$, explain why its solution set is not a subspace. Is containing the zero vector alone sufficient for a set to be a subspace? Use Exercise 1 to justify your answer.

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## B. Echelon form and nullspaces

5. Use row operations to find the reduced row echelon form of

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 8 \\ 1 & 2 & 2 & 5 \end{pmatrix}.$$

State its rank, pivot columns, basic variables, and free variables. Keep this matrix $A$ for Exercises 6, 7, 9, 10, 17, and 18.

6. For the matrix $A$ in Exercise 5, solve $A\mathbf{x} = \mathbf{0}$. Give a basis and the dimension of $N(A)$. Put your basis vectors into a matrix $N$ and verify $AN = 0$. Describe the nullspace geometrically.

7. Let $R = \text{rref}(A)$ for the matrix in Exercise 5. Interchange columns 2 and 3 using a permutation matrix $Q$ so that $RQ = \begin{pmatrix} I_2 & F \\ 0 & 0 \end{pmatrix}$. Find $Q$ and $F$. Use

$$N = Q \begin{pmatrix} -F \\ I_2 \end{pmatrix}$$

to obtain a nullspace basis in the original variable order. Explain why omitting $Q$ gives the wrong variable order in this example.

8. Find the rref, rank, and a basis for the nullspace of

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & -1 & 3 \end{pmatrix}.$$

Explain why a homogeneous system can have infinitely many solutions even when there are more equations than unknowns. Retain $B$ for Exercises 19 and 20.

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## C. Consistency and complete solution sets

9. For the matrix $A$ in Exercise 5, use elimination on $[A \mid \mathbf{b}]$ to find the condition on $\mathbf{b} = (b_1, b_2, b_3)^T$ for consistency. Express $C(A)$ as a plane in $\mathbb{R}^3$. Test the right-hand sides $(2, 5, 3)^T$ and $(2, 5, 4)^T$.

10. Solve $A\mathbf{x} = (2, 5, 3)^T$ for the matrix in Exercise 5. Find a particular solution by setting the free variables to zero, and then give every solution as $\mathbf{x}_p + \mathbf{x}_n$. Verify the particular solution and describe the solution set geometrically.

11. For each matrix size and rank below, state (i) whether $A\mathbf{x} = \mathbf{b}$ is solvable for every right-hand side, (ii) the possible numbers of solutions as $\mathbf{b}$ varies, and (iii) the dimensions of $N(A)$ and $N(A^T)$:

    (a) $4 \times 3,\ r = 3$; (b) $3 \times 5,\ r = 3$; (c) $4 \times 5,\ r = 2$.

    Give a brief rank-based justification for each case.

12. For real parameters $a$ and $\beta$, classify the system

$$x + y = 1, \qquad y + z = 2, \qquad x + 2y + az = \beta$$

    as having a unique solution, infinitely many solutions, or no solution. Give the complete solution whenever one exists.

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## D. Independence, spanning sets, and bases

13. For which real values of $a$ are

$$\mathbf{v}_1 = (1,0,1)^T, \qquad \mathbf{v}_2 = (0,1,1)^T, \qquad \mathbf{v}_3 = (1,1,a)^T$$

linearly independent? Use elimination rather than determinants. When they are dependent, give a nontrivial dependence relation and describe their span.

14. Let $\mathbf{u} = (1,0,1)^T$, $\mathbf{v} = (0,1,1)^T$, $\mathbf{w} = (1,1,2)^T$, and $\mathbf{q} = (2,-1,1)^T$. Find the dimension of their span and give two different bases chosen from these four vectors. Express $(3,-1,2)^T$ as a combination of each chosen basis.

15. Set $a = 3$ in Exercise 13. Prove that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$, then express $(2,1,4)^T$ as a linear combination of these vectors. Explain why this representation is unique.

16. Decide whether each statement is true or false. Prove true statements and give a counterexample to false statements.

    (a) Any set of four vectors in $\mathbb{R}^3$ is dependent.

    (b) Row operations preserve the column space of every matrix.

    (c) If $A$ is $m \times n$ of rank $r$, then $\dim N(A) = m - r$.

    (d) An independent set of $n$ vectors in $\mathbb{R}^n$ is a basis.

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## E. The four fundamental subspaces

17. For the matrix $A$ in Exercise 5, give a basis, dimension, and ambient space for each of $C(A)$, $N(A)$, $C(A^T)$, and $N(A^T)$. Use original pivot columns for the column-space basis and nonzero rref rows for the row-space basis.

18. Append $I_3$ to the matrix $A$ in Exercise 5. Apply the same row operations as in Exercise 5 to both blocks, reducing the coefficient block to $R = \text{rref}(A)$. Write the result as $[R \mid G]$ and verify $GA = R$. Use the row of $G$ corresponding to the zero row of $R$ to find a left-nullspace basis. Do not continue reducing by using pivots in the right block.

19. For the matrix $B$ in Exercise 8, find bases for all four fundamental subspaces. Use the left nullspace to derive all consistency conditions for $B\mathbf{x} = \mathbf{b}$. Then give the complete solution for $\mathbf{b} = (1, 2, 3, 0)^T$.

20. Show that consistency of $A\mathbf{x} = \mathbf{b}$ implies $\mathbf{y}^T\mathbf{b} = 0$ for every $\mathbf{y} \in N(A^T)$. Explain, using row reduction and the zero coefficient rows, why these conditions are also sufficient. Apply the criterion to $B\mathbf{x} = (1, 2, 3, 1)^T$, where $B$ is from Exercise 8.